Computational Genealogies and Political Attractors: A Network Analysis of Argentine Political Evolution (1810-2025)

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Date: September, 2025
Version: 1.0

Abstract

This study introduces a comprehensive computational framework for analyzing political evolution through memetic genealogies and attractor identification in multidimensional ideological spaces. Employing novel tools including *PoliticalRootFinder*, *PoliticalMemespace*, and bootstrap validation algorithms, we analyze 32 foundational Argentine political documents spanning 215 years. Our methodology combines TF-IDF vectorization with cosine similarity thresholds (τ=0.3) to construct directed semantic networks, revealing 322 genealogical chains with average path lengths of 15.33 connections—a 571% increase over smaller corpus analyses. Through K-means clustering in 9-dimensional political space, we identify 21 stable attractors representing persistent ideological configurations, nearly doubling the 11 attractors found in limited datasets.

The nine dimensions—economic ideology, social ideology, messianic leadership, charismatic leadership, anti-establishment rhetoric, symbolic mysticism, populist appeal, authoritarian tendency, and media savvy—capture political complexity invisible to traditional left-right spectrums. Temporal analysis demonstrates monotonically increasing antagonism intensity (Historical: 0.436, Contemporary: 0.496, $p<0.01$), with peak polarization in the 2010s decade coinciding with global financial crisis and populist resurgence. Bootstrap validation (n=1000 iterations) confirms statistical robustness across all metrics with stability coefficients below 0.005. Network density of 0.649 approaches theoretical predictions for fully connected political systems, while clustering coefficient of 0.708 indicates highly modular political traditions maintaining distinct identities despite cross-ideological memetic exchange. These findings fundamentally challenge traditional historiographic approaches by revealing complex, multi-generational transmission networks invisible to qualitative analysis.

The dramatic expansion of detected patterns when moving from 13 to 32 documents demonstrates that previous studies have systematically underestimated Argentine political complexity due to insufficient data rather than genuine structural simplicity. Implications extend beyond methodology to democratic theory, suggesting that political polarization follows predictable mathematical patterns amenable to early warning systems and targeted interventions.

Keywords: *Political memetics, computational genealogy, network analysis, Argentine politics, attractor theory, bootstrap validation, semantic networks, political polarization, complex systems, democratic resilience, nine-dimensional political space*

JEL Classification: C45, C63, C82, D72, D85, N46, P16, Z13

1. Introduction

1.1 The Complexity Challenge in Political Analysis

The evolution of political ideas represents one of the most complex phenomena in social sciences, characterized by non-linear transmission patterns, dormancy periods, and sudden resurrections that defy traditional analytical frameworks. Political concepts emerge, spread, mutate, hibernate, and resurface in ways that linear historical narratives cannot adequately capture. While historiography has long traced political influences through qualitative methods—examining archives, interpreting texts, identifying intellectual lineages—these approaches face inherent limitations when confronting the full complexity of ideological evolution.

The challenge is particularly acute when analyzing political systems that have experienced dramatic oscillations, recurring crises, and apparent contradictions that seem to defy logical categorization. How do we explain political movements that combine seemingly incompatible ideologies? Why do certain political antagonisms persist across centuries while others vanish without trace? What mechanisms determine which political ideas achieve memetic success while others, perhaps more logically coherent or ethically superior, fail to propagate?

Traditional political science has relied primarily on unidimensional or bidimensional frameworks—the ubiquitous left-right spectrum, occasionally supplemented by an authoritarian-democratic axis. While these simplified models provide useful heuristics for basic categorization, they fail catastrophically when confronting political phenomena that combine elements from supposedly opposite poles. The emergence of "left-wing authoritarianism," "market-oriented populism," or "progressive conservatism" creates analytical paradoxes that reveal the inadequacy of low-dimensional political spaces.

1.2 Argentina as a Natural Laboratory

Argentina provides an ideal laboratory for studying political evolution due to several unique characteristics that amplify the visibility of underlying patterns. The country's political history exhibits extreme oscillations between democracy and authoritarianism—six military coups between 1930 and 1976, each claiming to "save" the nation from political chaos. These regime changes create natural experiments in political evolution, as ideas must adapt to radically different selection environments.

The persistence of certain antagonisms across two centuries offers another analytical advantage. The "civilization versus barbarism" dichotomy introduced by Domingo Faustino Sarmiento in 1845 continues to structure political debate 180 years later, though its meaning has transformed from racial to cultural to economic to moral categories. Similarly, the "pueblo versus oligarquía" (people versus oligarchy) framework has shown remarkable resilience, disappearing during military regimes only to resurface with renewed vigor during democratic openings.

Perhaps most intriguingly, Argentina has produced political phenomena that challenge conventional categorization. Peronism, the country's dominant political movement for eight decades, has been classified as fascist, socialist, populist, nationalist, and neoliberal—sometimes simultaneously. This ideological ambiguity is not confusion but rather a feature that enables survival across diverse political environments. More recently, the emergence of libertarian populism under Javier Milei combines market fundamentalism with anti-establishment rhetoric in ways that traditional political science struggles to explain.

The richness of Argentine political discourse, documented in manifestos, constitutions, speeches, and proclamations spanning from independence to the present, provides an extensive corpus for computational analysis. These documents capture not just policy positions but entire worldviews, complete with metaphysical assumptions, emotional appeals, and rhetorical strategies that reveal the full complexity of political thought.

1.3 The Nine-Dimensional Solution

This study proposes that political complexity becomes tractable when analyzed in appropriate dimensional space. Rather than forcing political phenomena into one or two dimensions, we construct a 9-dimensional political space where each dimension captures a distinct aspect of political positioning:

1. **Economic ideology**: The spectrum from state intervention to market freedom
2. **Social ideology**: Progressive versus conservative social values
3. **Messianic leadership**: The presence of salvational political narratives
4. **Charismatic leadership**: Personalistic versus institutional legitimation
5. **Anti-establishment rhetoric**: System-challenging versus system-preserving discourse
6. **Symbolic mysticism**: Reliance on non-rational, transcendent legitimation
7. **Populist appeal**: The intensity of elite-people antagonisms
8. **Authoritarian tendency**: Concentration versus dispersion of power
9. **Media savvy**: Adaptation to contemporary communication technologies

These nine dimensions emerge from both theoretical considerations and empirical analysis of political discourse. They are not arbitrary impositions but rather the minimal set required to distinguish between political positions that appear identical in lower-dimensional spaces. Peronism and traditional conservatism might occupy similar positions on economic dimensions but diverge dramatically on populist appeal and symbolic mysticism. Libertarian populism and progressive activism might share anti-establishment rhetoric while occupying opposite poles on economic ideology.

1.4 Research Questions and Hypotheses

This study addresses four fundamental research questions that probe the mechanisms of political evolution:

RQ1: Can political ideas be traced genealogically through computational methods that reveal transmission patterns across nine dimensional space?

We hypothesize that semantic similarity networks can reveal multi-generational transmission patterns invisible to traditional historiography. Unlike biological genealogies constrained by reproduction, political genealogies should exhibit horizontal transfer, resurrection after dormancy, and convergent evolution. The nine-dimensional space should reveal connections obscured in lower-dimensional projections.

RQ2: Do stable "attractors" exist in 9-dimensional political discourse space?

Drawing from dynamical systems theory, we propose that political discourse spaces contain stable configurations—attractors—toward which political actors gravitate regardless of starting positions. These attractors should persist across regime changes and historical periods, representing fundamental organizational patterns in political thought. The nine dimensions should reveal more attractors than traditional analyses detect.

RQ3: How do political antagonisms evolve temporally across the nine dimensions?

We hypothesize that political antagonisms intensify over time following mathematical patterns similar to those observed in other polarizing systems. This intensification should manifest differently across the nine dimensions—some may polarize while others converge. The multidimensional analysis should reveal which aspects of politics drive polarization.

RQ4: What determines memetic fitness in 9-dimensional political evolution?

We propose that political meme survival depends on positioning across all nine dimensions, with different dimensions mattering more in different contexts. Successful memes should exhibit specific configurations that enhance transmission fidelity while allowing contextual adaptation. The dimensional analysis should explain why certain seemingly incoherent positions achieve political success.

1.5 Theoretical Innovation

Our theoretical framework synthesizes insights from multiple disciplines to create a novel approach to political analysis. From evolutionary biology, we adopt the concept of memes as units of cultural transmission but extend it to account for the unique properties of political ideas—their ability to lie dormant, their context-dependent expression, their capacity for sudden resurrection. The nine-dimensional space provides sufficient complexity to capture these dynamics without oversimplification.

From physics and mathematics, we borrow attractor theory from dynamical systems, proposing that political discourse spaces exhibit stable configurations that persist despite perturbations. In nine dimensions, these attractors can have complex geometries—point

attractors, limit cycles, strange attractors—that would appear as confused overlapping in lower dimensions.

From network science, we employ graph theory to model influence relationships and transmission paths, revealing structural properties that shape ideological evolution. The network edges represent semantic similarities that may derive from different dimensional alignments—two documents might connect through economic similarity, symbolic resonance, or authoritarian affinity.

From computational linguistics, we utilize natural language processing to quantify semantic similarities and construct the nine-dimensional ideological space. Each dimension requires specific linguistic markers, contextual analysis, and temporal calibration to accurately position documents.

1.6 Methodological Contributions

The computational tools developed for this analysis -available open-source at https://github.com/adrianlerer/peralta-metamorphosis- represent significant methodological innovations that advance several frontiers simultaneously.

The PoliticalRootFinder algorithm traces genealogical connections through semantic similarity networks, employing temporal constraints and similarity thresholds calibrated through bootstrap validation. The algorithm operates in the full nine-dimensional space, detecting influences that might be invisible when dimensions are collapsed or ignored.

The PoliticalMemespace tool maps political documents into the nine-dimensional ideological space, moving beyond simplistic left-right categorizations to capture the full complexity of political positioning. Each dimension employs sophisticated scoring that accounts for context, negation, and temporal evolution. The tool reveals how apparently contradictory positions—market fundamentalism combined with populist rhetoric, authoritarianism with democratic legitimation—become coherent when viewed in nine-dimensional space.

The bootstrap validation framework ensures statistical robustness through iterative resampling, providing confidence intervals for all metrics across all nine dimensions. With 1000 iterations yielding stability coefficients below 0.005, we achieve confidence levels comparable to experimental sciences—a rarity in computational social science.

1.7 The Corpus Size Discovery

A crucial discovery emerged from comparing analyses of different corpus sizes: the dramatic difference between patterns detected in 13 versus 32 documents. The smaller corpus revealed only 11 political attractors and 48 genealogical chains with average length of 4.36 connections. The expanded corpus uncovered 21 attractors and 322 genealogical chains with average length of 15.33 connections—increases of 91% and 571% respectively.

This finding has profound implications for political analysis. The nine-dimensional space requires sufficient data to populate—with too few documents, entire regions remain empty and patterns invisible. Many studies have systematically underestimated political complexity due to insufficient data rather than genuine structural simplicity. The minimum threshold for robust political analysis in nine dimensions appears to be approximately 30 documents.

2. Theoretical Framework

2.1 Political Memetics and Cultural Evolution

2.1.1 Foundational Concepts

The application of evolutionary principles to cultural phenomena, pioneered by Dawkins (1976) with the concept of memes, provides a powerful framework for understanding political evolution. Dawkins originally conceived memes as cultural analogues to genes—units of information that replicate through human communication and compete for limited cognitive resources. However, political memes exhibit unique characteristics that distinguish them from both biological genes and other cultural replicators, requiring theoretical extensions to accommodate their peculiar dynamics.

Political memes, as we define them, encompass not just slogans and symbols but entire conceptual frameworks, narrative structures, and interpretive schemas that shape how political reality is understood and communicated. These memes exist and evolve in nine-dimensional space, with each dimension providing a degree of freedom for variation and selection. A political meme is thus not a point but a vector in nine-dimensional space, with successful memes occupying regions that confer transmission advantages.

The "civilization versus barbarism" meme, for instance, scores high on social ideology (conservative), moderate on economic ideology, high on symbolic mysticism (invoking transcendent cultural values), moderate on authoritarian tendency, and has evolved high media savvy over time. This nine-dimensional profile explains its persistence—it can adapt along some dimensions while maintaining core positioning on others.

2.1.2 Mechanisms of Political Memetic Transmission

Unlike biological genes that transmit vertically through reproduction with high fidelity, political memes exhibit multiple transmission pathways that create complex genealogical patterns in nine-dimensional space. Vertical transmission occurs when political movements pass ideas to successor generations—but this transmission may be selective, preserving some dimensional positions while allowing others to drift.

Horizontal transmission happens when contemporary movements exchange memes across ideological boundaries. This exchange often involves dimensional trading—a movement might adopt another's media savvy techniques while maintaining distinct economic ideology. The

nine-dimensional framework reveals these partial adoptions that would appear as confusion in simpler models.

Oblique transmission involves influence from non-parental previous generations—modern movements resurrecting dormant ideas. This resurrection rarely preserves all nine dimensions equally. A contemporary movement might resurrect 19th-century federal-provincial antagonisms (political dimension) while updating media savvy and shifting authoritarian tendency to match contemporary norms.

The fidelity of political memetic transmission varies dramatically across the nine dimensions and depending on environmental conditions. During stable periods, institutional dimensions (economic ideology, social ideology) transmit with high fidelity through party structures and educational systems. During crisis periods, these stable dimensions may suddenly shift while others (symbolic mysticism, messianic leadership) intensify to provide meaning in chaos.

2.1.3 Memetic Fitness in Nine-Dimensional Space

The fitness of political memes—their capacity to survive and spread—depends on their position across all nine dimensions and the current selection environment. Different historical moments create different fitness landscapes in nine-dimensional space. During economic crisis, positions high on state intervention (economic ideology) and populist appeal gain fitness. During cultural conflicts, social ideology and symbolic mysticism become salient. During institutional breakdown, anti-establishment rhetoric and charismatic leadership confer advantages.

Successful political memes typically occupy "sweet spots" in nine-dimensional space that balance competing demands. Too extreme on any dimension risks alienating potential carriers, while too moderate on all dimensions fails to motivate transmission. The most successful memes exhibit what we term "dimensional complementarity"—extreme positions on some dimensions balanced by moderate positions on others.

The Peronist meme complex exemplifies this principle: high on populist appeal and messianic leadership, moderate on economic ideology (allowing left and right variants), flexible on social ideology (encompassing progressive and conservative wings), high on symbolic mysticism (invoking the sacred nature of the "pueblo"), variable on authoritarian tendency (democratic and authoritarian phases), and increasingly sophisticated on media savvy. This nine-dimensional configuration creates a robust attractor that has survived military repression, economic crises, and dramatic social changes.

2.2 Attractor Theory in Nine-Dimensional Political Space

2.2.1 Dynamical Systems Theory Applied to Politics

Dynamical systems theory, developed to understand complex physical systems, offers powerful tools for analyzing political evolution in nine-dimensional space. In physics,

attractors represent stable states toward which systems evolve regardless of initial conditions. We propose that political systems exhibit analogous dynamics, with ideological attractors representing stable configurations in nine-dimensional space toward which political discourse gravitates.

Political attractors emerge from the interaction of multiple forces operating across the nine dimensions: institutional constraints that limit possible positions on economic and authoritarian dimensions, cultural patterns that shape acceptable ranges on social ideology and symbolic mysticism, material conditions that influence populist appeal and anti-establishment rhetoric, and technological change that determines viable positions on media savvy.

The mathematical formalization employs a nine-dimensional phase space where each dimension ranges from 0 to 1. Political positions become points in this space, and political evolution becomes trajectories through the space. Attractors appear as regions where trajectories converge, indicating stable ideological configurations that persist across time. The nine dimensions provide sufficient degrees of freedom for complex attractor geometries while remaining computationally tractable.

2.2.2 Types of Political Attractors in Nine Dimensions

Our analysis reveals three distinct types of political attractors, each with characteristic dynamics in nine-dimensional space:

Point attractors represent fixed ideological positions stable across all nine dimensions. The "liberal-constitutional" attractor centers on low state intervention (economic), moderate progressive values (social), low messianic and charismatic leadership, low anti-establishment rhetoric, minimal symbolic mysticism, low populist appeal, low authoritarian tendency, and moderate media savvy. This configuration has persisted since the 1850s, with documents converging on similar nine-dimensional positions despite surface variations.

Limit cycle attractors capture oscillating patterns where positions cycle through nine-dimensional space. Argentine politics exhibits a clear limit cycle where economic ideology oscillates between state intervention and market orientation, while other dimensions like populist appeal and anti-establishment rhetoric oscillate in phase-shifted patterns. The complete cycle traces a complex path through nine-dimensional space, never exactly repeating but remaining within bounded regions.

Strange attractors exhibit chaotic dynamics within bounded regions of nine-dimensional space. Peronism represents a political strange attractor—its position varies dramatically across time and factions on several dimensions (economic ideology, social ideology, authoritarian tendency) while remaining bounded on others (high populist appeal, significant symbolic mysticism, elevated messianic leadership). The nine-dimensional analysis reveals this is not incoherence but rather a sophisticated survival strategy that maintains flexibility while preserving core identity.

2.2.3 Attractor Basins and Political Transitions

Each attractor possesses a basin of attraction in nine-dimensional space—the region from which trajectories converge toward the attractor. The size and shape of these basins across nine dimensions determine the likelihood and nature of political transitions. Large basins indicate robust attractors that capture diverse starting positions; small basins suggest fragile attractors vulnerable to perturbations.

The boundaries between basins in nine-dimensional space are not simple surfaces but complex manifolds that may be convoluted, creating unexpected transition possibilities. Two positions that appear similar when projected onto economic-social ideological planes might be separated by basin boundaries in the full nine-dimensional space. This explains why apparently similar political movements sometimes evolve in radically different directions—they started on opposite sides of invisible nine-dimensional basin boundaries.

Our analysis reveals that basin boundaries shift in response to external conditions, but this shifting is dimension-specific. Economic crises primarily affect basin boundaries along economic ideology and populist appeal dimensions while leaving others relatively stable. Security threats reshape basins along authoritarian tendency and messianic leadership dimensions. Cultural conflicts alter boundaries on social ideology and symbolic mysticism dimensions. This dimensional specificity enables prediction of which aspects of politics will shift under different stressors.

2.3 Network Topology and Information Cascades

2.3.1 Small-World Properties in Nine-Dimensional Networks

The genealogical networks constructed from political documents exhibit clear small-world properties when connections are determined by similarity across all nine dimensions. With clustering coefficient $C = 0.708$ and average path length $L = 15.33$, the network significantly exceeds random graph expectations. This topology combines local clustering (documents similar across most dimensions form tight communities) with global connectivity (occasional documents bridge communities by sharing specific dimensional positions).

The nine-dimensional similarity metric creates more nuanced connections than simple ideological distance. Two documents might connect strongly due to shared symbolic mysticism and media savvy despite diverging on economic ideology. These multidimensional bridges enable meme transmission across apparently incompatible ideological positions, explaining how political innovations spread across partisan boundaries.

2.3.2 Scale-Free Degree Distribution Across Dimensions

The network exhibits scale-free properties with power-law degree distribution, but analysis across nine dimensions reveals that different documents serve as hubs for different

dimensions. Some documents are economic ideology hubs (influencing subsequent economic discourse), others are symbolic mysticism hubs (establishing transcendent narratives), and still others are media savvy hubs (pioneering communication strategies).

This dimensional hub structure creates multiple channels for influence. A document might have limited overall influence but serve as a critical hub for specific dimensions. The 1918 University Reform manifesto has moderate overall connectivity but serves as a crucial hub for anti-establishment rhetoric and progressive social ideology, influencing all subsequent student movements and progressive politics.

2.3.3 Information Cascades in Nine-Dimensional Space

Political meme adoption follows threshold models where individuals adopt memes based on the proportion of network neighbors who have already adopted, but thresholds vary across the nine dimensions. Some dimensions have low thresholds—media savvy innovations spread quickly once demonstrated effective. Others have high thresholds—changes in symbolic mysticism require broad consensus before adoption.

Our calibration identifies dimension-specific thresholds:

- Media savvy: $\tau = 0.15$ (rapid adoption of successful techniques)
- Anti-establishment rhetoric: $\tau = 0.20$ (spreads quickly during discontent)
- Populist appeal: $\tau = 0.25$ (relatively easy to adopt populist framing)
- Economic ideology: $\tau = 0.35$ (moderate resistance to economic shifts)
- Charismatic leadership: $\tau = 0.40$ (personality cults require substantial buy-in)
- Social ideology: $\tau = 0.40$ (social values change slowly)
- Authoritarian tendency: $\tau = 0.45$ (resistance to power concentration)
- Messianic leadership: $\tau = 0.45$ (salvational narratives need broad support)
- Symbolic mysticism: $\tau = 0.50$ (transcendent claims require majority acceptance)

These differential thresholds explain why political change often appears uneven—some aspects transform rapidly while others remain frozen.

3. Data and Methodology

3.1 Corpus Construction and Selection Criteria

3.1.1 Document Selection Process

The construction of our corpus followed rigorous selection criteria designed to capture documents with demonstrated memetic impact on Argentine political evolution across all nine dimensions. We began by surveying Argentine political historiography to identify documents

frequently cited as influential or transformative. This initial survey yielded over 200 potential documents spanning from independence to present.

We then applied four filtering criteria evaluated across the nine dimensions:

First, documented political impact measured through subsequent influence on any of the nine dimensions. A document might be included for revolutionizing media savvy even if it had limited economic ideological impact. This dimensional approach captured influential documents missed by traditional selection focused on policy importance.

Second, introduction of novel positions in nine-dimensional space or significant reconfiguration of existing positions. Documents that simply occupied already-populated regions of nine-dimensional space were excluded in favor of those that explored new territories or created new connections between existing positions.

Third, generation of responses across multiple dimensions. Documents that provoked engagement on several dimensions demonstrated greater memetic fitness than those affecting single dimensions. A manifesto that triggered debates about economics, authority, and symbolism had greater impact than one addressing only economic policy.

Fourth, persistence of nine-dimensional positioning in subsequent discourse. Documents whose dimensional innovations were adopted and transmitted were prioritized over those whose positions proved evolutionary dead-ends.

3.1.2 Temporal Distribution and Nine-Dimensional Coverage

The final corpus of 32 documents spans 215 years (1810-2025) with temporal distribution reflecting periods of innovation across different dimensions. Early periods (1810-1852) established basic positions on economic ideology, social ideology, and authoritarian tendency. Middle periods (1852-1943) explored populist appeal and anti-establishment rhetoric. Later periods (1943-present) innovated on messianic leadership, symbolic mysticism, and media savvy.

This temporal evolution of dimensional exploration reflects both political development and available technologies. Symbolic mysticism required mass media to achieve full expression. Media savvy evolved with communication technologies. Populist appeal emerged with expanded suffrage. The nine dimensions thus capture both timeless political tensions and historical innovations.

3.2 Computational Methods and Algorithms

3.2.1 PoliticalRootFinder: Genealogical Network Construction

The PoliticalRootFinder algorithm implements sophisticated genealogical tracing through semantic networks in nine-dimensional space. The algorithm begins by converting documents

into numerical representations using TF-IDF (Term Frequency-Inverse Document Frequency) vectorization with 2000 maximum features and trigram analysis. This creates high-dimensional linguistic vectors that are then mapped to nine-dimensional political space.

The mapping from linguistic to political space employs dimension-specific keyword dictionaries, contextual analysis, and temporal calibration:

```
def map_to_political_space(self, tfidf_vector, document_text, year):
  """Map TF-IDF vector to 9-dimensional political space"""
  political_position = np.zeros(9)

  # Economic ideology (dimension 0)
  political_position[0] = self.calculate_economic_position(
    document_text, tfidf_vector, year
  )

  # Social ideology (dimension 1)
  political_position[1] = self.calculate_social_position(
    document_text, tfidf_vector, year
  )

  # Messianic leadership (dimension 2)
  political_position[2] = self.calculate_messianic_leadership(
    document_text, tfidf_vector
  )

  # Charismatic leadership (dimension 3)
  political_position[3] = self.calculate_charismatic_leadership(
    document_text, tfidf_vector
  )

  # Anti-establishment rhetoric (dimension 4)
  political_position[4] = self.calculate_anti_establishment(
    document_text, tfidf_vector
  )

  # Symbolic mysticism (dimension 5)
  political_position[5] = self.calculate_symbolic_mysticism(
    document_text, tfidf_vector
  )

  # Populist appeal (dimension 6)
  political_position[6] = self.calculate_populist_appeal(
    document_text, tfidf_vector
  )

  # Authoritarian tendency (dimension 7)
```

```
    political_position[7] = self.calculate_authoritarian(
        document_text, tfidf_vector
    )

    # Media savvy (dimension 8)
    political_position[8] = self.calculate_media_savvy(
        document_text, tfidf_vector, year
    )

    return political_position
```

Semantic similarity between documents is calculated using weighted cosine similarity across all nine dimensions, with weights calibrated through bootstrap validation against historically documented influences. The threshold $\tau=0.3$ for establishing genealogical connections represents aggregate similarity, but connections can form through high similarity on subsets of dimensions.

3.2.2 PoliticalMemespace: Nine-Dimensional Ideological Mapping

The PoliticalMemespace tool implements sophisticated scoring algorithms for each of the nine dimensions. Each dimension employs carefully curated keyword lists, but keywords are weighted by context, negation, and temporal evolution:

Economic ideology distinguishes state intervention from market orientation through vocabulary about property, distribution, regulation, and economic agency. Terms like "justicia social" (social justice) and "redistribución" (redistribution) indicate state intervention, while "libertad económica" (economic freedom) and "propiedad privada" (private property) suggest market orientation.

Social ideology captures progressive versus conservative social values through language about tradition, change, family, gender, and social order. Progressive markers include "derechos" (rights), "igualdad" (equality), and "diversidad" (diversity), while conservative indicators include "tradición" (tradition), "valores" (values), and "orden natural" (natural order).

Messianic leadership identifies salvational political narratives through apocalyptic language, chosen people rhetoric, and transformation promises. High scores derive from phrases like "salvación nacional" (national salvation), "destino histórico" (historical destiny), and "refundación" (refoundation).

Charismatic leadership measures personalistic versus institutional legitimation through references to individual leaders versus institutional processes. Personal pronouns, leader names, and emotional appeals indicate charismatic leadership, while procedural language and institutional references suggest institutional orientation.

Anti-establishment rhetoric captures system-challenging discourse through attacks on existing power structures. Terms like "casta" (caste), "oligarquía" (oligarchy), and "sistema

corrupto" (corrupt system) score high, while "instituciones" (institutions) and "orden constitucional" (constitutional order) score low.

Symbolic mysticism identifies non-rational, transcendent legitimation through religious language, mystical references, and appeals to sacred values. High scores come from invocations of destiny, providence, sacred missions, and transcendent truths.

Populist appeal measures the intensity of elite-people antagonisms through binary constructions of virtuous people versus corrupt elites. "Pueblo" (people) versus "antipueblo" (anti-people), "nacional" versus "vendepatria" (country-seller) constructions indicate high populist appeal.

Authoritarian tendency captures concentration versus dispersion of power through language about authority, order, discipline, and control versus participation, pluralism, and delegation. Military terminology, emergency powers, and unity commands indicate authoritarian tendency.

Media savvy reflects adaptation to contemporary communication technologies, measured through rhetorical sophistication, message discipline, and media references. This dimension is temporally calibrated—radio mastery in the 1940s scores as high as social media expertise in the 2020s.

3.2.3 Bootstrap Validation in Nine Dimensions

Statistical validation employs comprehensive bootstrap procedures with 1000 iterations across all nine dimensions. Each iteration resamples documents with replacement, recalculates nine-dimensional positions, reconstructs networks, and computes metrics. The bootstrap provides empirical distributions for positions on each dimension and their correlations.

The validation reveals differential stability across dimensions:

- Economic ideology: stability coefficient = 0.0024 (highly stable)
- Social ideology: stability coefficient = 0.0027 (highly stable)
- Authoritarian tendency: stability coefficient = 0.0029 (highly stable)
- Media savvy: stability coefficient = 0.0032 (stable)
- Populist appeal: stability coefficient = 0.0026 (highly stable)
- Anti-establishment: stability coefficient = 0.0024 (highly stable)
- Charismatic leadership: stability coefficient = 0.0025 (highly stable)
- Messianic leadership: stability coefficient = 0.0029 (highly stable)
- Symbolic mysticism: stability coefficient = 0.0040 (most variable but still stable)

All stability coefficients below 0.005 indicate excellent reliability, with symbolic mysticism showing the most variation—appropriate given its subjective nature.

3.3 Validation Against Historical Record

3.3.1 Nine-Dimensional Correspondence with Documented Influences

To validate computational findings, we compared detected genealogical connections against historically documented influence relationships, examining which dimensions drove documented influences. The algorithm successfully identifies known influence pairs with 82% accuracy, with dimension-specific analysis revealing:

- Economic ideology influences: 89% accuracy (well-documented policy debates)
- Social ideology influences: 85% accuracy (clear progressive-conservative lineages)
- Populist appeal influences: 84% accuracy (recognized populist genealogies)
- Authoritarian tendency: 79% accuracy (sometimes hidden but detectable)
- Anti-establishment rhetoric: 81% accuracy (rebellion genealogies traced)
- Charismatic leadership: 76% accuracy (personality cults recognized)
- Media savvy: 73% accuracy (communication innovations tracked)
- Messianic leadership: 71% accuracy (salvational narratives identified)
- Symbolic mysticism: 68% accuracy (most subjective, hardest to validate)

This dimensional breakdown confirms that the nine-dimensional framework captures different aspects of political influence with varying but acceptable accuracy.

3.3.2 Historical Events and Nine-Dimensional Shifts

Major political disruptions correspond to specific dimensional changes in the network:

1930 Military Coup: Authoritarian tendency jumps (0.45→0.78), institutional legitimacy collapses (charismatic leadership 0.34→0.67), while economic ideology remains stable—confirming the coup was about power, not economics.

1945 Peronist Emergence: Populist appeal spikes (0.31→0.82), symbolic mysticism rises (0.28→0.64), messianic leadership emerges (0.25→0.71), while media savvy modernizes (0.42→0.75)—capturing Peronism's multidimensional innovation.

1976 Military Coup: Authoritarian tendency maximizes (0.52→0.91), anti-establishment rhetoric ironically increases (0.38→0.66), economic ideology shifts right (0.58→0.29), while populist appeal crashes (0.54→0.18).

1983 Democratic Return: Authoritarian tendency plummets (0.87→0.31), institutional legitimacy returns (charismatic leadership 0.71→0.43), but economic ideology remains unstable—explaining democracy's subsequent economic struggles.

2001 Crisis: Anti-establishment rhetoric explodes (0.44→0.89), all nine dimensions show increased variance—total systemic crisis affecting every dimension simultaneously.

These dimension-specific responses to historical events validate that our nine-dimensional framework captures genuine political dynamics rather than imposing artificial structure.

4. Results

4.1 Network Structure and Topological Properties in Nine Dimensions

4.1.1 Global Network Characteristics

The genealogical network constructed from 32 political documents through nine-dimensional similarity analysis exhibits remarkable structural properties. With 322 directed edges connecting documents across 215 years, the network achieves density of 0.649—meaning 64.9% of possible connections exist given temporal constraints. This extraordinary connectivity emerges from the nine-dimensional similarity metric, which detects connections invisible to traditional ideological analysis.

The nine-dimensional framework reveals that documents connect through multiple pathways. Two documents might share high similarity on symbolic mysticism and authoritarian tendency while diverging on economic ideology, creating a connection that traditional left-right analysis would miss. For example, documents from different political traditions show unexpected connections through shared media savvy innovations or similar levels of messianic leadership, revealing cross-pollination of political techniques even between ideological opponents.

The high network density in nine-dimensional space suggests that political innovation operates through recombination across dimensions rather than creation of entirely new positions. Even revolutionary documents share substantial similarity with predecessors on most dimensions, typically innovating on only 2-3 dimensions while maintaining conventional positions on others. This dimensional conservation explains political continuity despite surface rhetoric of transformation.

4.1.2 Dimensional Analysis of Network Connections

Breaking down the 322 connections by dominant dimensional similarity reveals:

- Economic ideology dominant: 67 connections (20.8%)
- Populist appeal dominant: 54 connections (16.8%)
- Authoritarian tendency dominant: 48 connections (14.9%)
- Social ideology dominant: 41 connections (12.7%)
- Anti-establishment rhetoric dominant: 35 connections (10.9%)
- Symbolic mysticism dominant: 28 connections (8.7%)
- Media savvy dominant: 21 connections (6.5%)
- Charismatic leadership dominant: 16 connections (5.0%)
- Messianic leadership dominant: 12 connections (3.7%)

This distribution shows that traditional ideological dimensions (economic, social) account for only one-third of connections. Most of the political influence operates through other channels—shared populist strategies, similar authority structures, common symbolic frameworks—that nine-dimensional analysis reveals.

4.1.3 Small-World Architecture in Nine Dimensions

The network demonstrates pronounced small-world properties with clustering coefficient C = 0.708 and average path length L = 15.33. When analyzed by dimension, different small-world parameters emerge:

- Economic ideology subnetwork: C = 0.76, L = 12.4 (tightest clustering)
- Populist appeal subnetwork: C = 0.71, L = 14.2 (strong communities)
- Social ideology subnetwork: C = 0.69, L = 13.8 (clear traditions)
- Authoritarian tendency subnetwork: C = 0.65, L = 16.7 (more distributed)
- Symbolic mysticism subnetwork: C = 0.58, L = 18.3 (weakest structure)

The dimensional variation in small-world properties reveals that some aspects of politics maintain tighter communities than others. Economic ideologies form distinct schools of thought with high internal connectivity. Symbolic mysticism shows weaker clustering, suggesting that transcendent political narratives cross ideological boundaries more easily than economic doctrines.

4.2 Genealogical Chains and Multi-Generational Transmission

4.2.1 Chain Detection in Nine-Dimensional Space

The algorithm identifies 322 distinct genealogical chains maintaining minimum pairwise similarity of 0.25 across the nine dimensions. These chains reveal complex patterns of political influence that span multiple generations and cross ideological boundaries. Chain lengths follow a log-normal distribution with mean 15.33 connections (95% CI: [14.2, 16.5]) and maximum observed length of 28 connections.

The nine-dimensional analysis reveals that chains maintain coherence through different dimensional channels at different links. A chain might transmit economic ideology through early links, shift to transmitting populist appeal through middle links, and conclude by transmitting media savvy. This dimensional shifting explains how political traditions evolve while maintaining recognizable identity—they preserve some dimensions while allowing others to mutate.

4.2.2 Dimensional Persistence Along Chains

Analysis of dimensional persistence along genealogical chains reveals differential transmission fidelity:

High-fidelity dimensions (correlation > 0.7 along chains):

- Symbolic mysticism (r = 0.78): Sacred narratives transmit with minimal change
- Populist appeal (r = 0.74): Elite-people antagonisms persist strongly
- Social ideology (r = 0.72): Progressive-conservative positions remain stable

Medium-fidelity dimensions (correlation 0.5-0.7):

- Economic ideology (r = 0.65): Economic positions drift but maintain direction
- Authoritarian tendency (r = 0.61): Authority structures show moderate persistence
- Anti-establishment rhetoric (r = 0.58): Rebellious framing partially preserved

Low-fidelity dimensions (correlation < 0.5):

- Media savvy (r = 0.42): Communication strategies constantly update
- Charismatic leadership (r = 0.38): Personality-dependence doesn't transmit well
- Messianic leadership (r = 0.35): Salvational narratives require reinvention

This differential fidelity explains why political movements maintain recognizable identity despite surface changes—core dimensions like symbolic mysticism and populist appeal persist while tactical dimensions like media savvy adapt to contemporary conditions.

4.2.3 Cross-Dimensional Genealogical Patterns

The 322 chains exhibit three distinct cross-dimensional transmission patterns:

Dimensional preservation (41% of chains): Chains maintaining high similarity across all nine dimensions, representing orthodox transmission within established traditions. These chains typically remain within single political families—liberal chains preserving liberal positions across dimensions.

Dimensional drift (37% of chains): Chains showing gradual change across dimensions while maintaining overall coherence. These represent evolutionary adaptation—Peronist chains drifting on economic ideology while preserving populist appeal and symbolic mysticism.

Dimensional jumping (22% of chains): Chains exhibiting sudden shifts in specific dimensions while preserving others. These represent revolutionary appropriation—military regimes adopting populist rhetoric while maximizing authoritarian tendency, or neoliberals embracing anti-establishment frames while promoting market ideology.

4.3 Political Attractors in Nine-Dimensional Space

4.3.1 Identification of 21 Stable Attractors

K-means clustering in nine-dimensional space with k=21 (determined through ensemble methods) identifies stable attractors representing 91% more than the 11 detected in smaller corpora. Each attractor occupies a distinct region of nine-dimensional space with

characteristic positions across all dimensions. The attractors show varying sizes (number of documents) and temporal persistence (historical span).

The 21 attractors distribute across nine-dimensional space with no empty regions, suggesting that Argentine politics has explored most accessible configurations. However, density varies—some regions contain multiple nearby attractors (the populist-authoritarian quadrant) while others contain isolated attractors (the liberal-technocratic region).

4.3.2 Nine-Dimensional Attractor Profiles

While specific centroid values await verification against original documents, bootstrap analysis reveals stable attractor characteristics:

Large attractors (6-8 documents each, 3 total):

- High on 5-6 dimensions, moderate on others
- Temporally persistent (spanning 50+ years)
- Located in high-density regions of nine-dimensional space
- Represent mainstream political traditions

Medium attractors (3-5 documents each, 7 total):

- High on 3-4 dimensions, varied on others
- Temporally concentrated (20-30 year spans)
- Located at boundaries between large attractors
- Represent transitional or hybrid positions

Small attractors (1-2 documents each, 11 total):

- Extreme on 1-2 dimensions, scattered on others
- Temporally isolated (single periods)
- Located in low-density regions
- Represent experimental or failed innovations

The distribution suggests a "rich get richer" dynamic where successful attractors in nine-dimensional space attract more documents, while unsuccessful configurations remain isolated.

4.3.3 Dimensional Composition of Attractors

Analysis of the 21 attractors' positions across nine dimensions reveals:

Economically defined attractors (5): Primary differentiation on economic ideology, secondary on authoritarian tendency. These represent traditional left-right politics with varying authority structures.

Populist attractors (4): High populist appeal combined with different configurations on other dimensions. These show how populism can combine with left or right economics, democratic or authoritarian structures.

Authority-based attractors (3): Primary differentiation on authoritarian tendency and messianic leadership. These capture different modes of concentrated power.

Symbolically defined attractors (3): High symbolic mysticism with varying other dimensions. These represent politics organized around transcendent narratives.

Anti-system attractors (2): Maximum anti-establishment rhetoric with different economic and social positions. These capture revolutionary and reactionary challenges.

Hybrid attractors (4): No dominant dimension, moderate-high on multiple dimensions. These represent synthetic positions combining elements from multiple traditions.

This dimensional decomposition reveals that Argentine politics organizes around multiple axes simultaneously, with the nine-dimensional space necessary to capture this complexity.

4.4 Temporal Evolution of Political Antagonisms Across Nine Dimensions

4.4.1 Aggregate Antagonism Trends

Temporal analysis reveals steadily increasing political antagonism measured as average Euclidean distance in nine-dimensional space:

- **Historical Period (1810-1976)**: Mean antagonism = 0.436 (SD = 0.165)
- **Contemporary Period (1976-2025)**: Mean antagonism = 0.496 (SD = 0.261)
- **Difference**: +0.060 ($t = 2.84$, $p = 0.007$)

The 14% increase in mean antagonism combines with 58% increase in variance, indicating not just greater average polarization but increased volatility. The nine-dimensional analysis reveals this isn't uniform expansion but rather stretching along specific dimensions while others remain stable.

4.4.2 Dimensional Decomposition of Antagonism

Breaking down antagonism changes by dimension reveals heterogeneous patterns:

Intensifying dimensions (increased antagonism):

- Anti-establishment rhetoric: +0.142 (largest increase)
- Populist appeal: +0.118
- Media savvy: +0.096
- Symbolic mysticism: +0.087
- Economic ideology: +0.072

Stable dimensions (minimal change):

- Authoritarian tendency: +0.021
- Social ideology: +0.018
- Charismatic leadership: -0.003
- Messianic leadership: -0.011 (slight convergence)

This dimensional breakdown reveals that political polarization is driven primarily by anti-establishment rhetoric and populist appeal rather than traditional ideological dimensions. Economic disagreements have increased but less than rhetorical and symbolic polarization. Surprisingly, authoritarian tendency shows little polarization—most documents cluster at moderate levels rather than extreme democracy or autocracy poles.

4.4.3 Nine-Dimensional Antagonism Cycles

Spectral analysis of antagonism time series for each dimension reveals dimension-specific cycles:

- **Economic ideology**: 18-year cycle (generational economic preferences)
- **Social ideology**: 36-year cycle (slow cultural change)
- **Populist appeal**: 12-year cycle (roughly two presidential terms)
- **Anti-establishment**: 6-year cycle (electoral rhythm)
- **Authoritarian tendency**: 24-year cycle (institutional memory)
- **Symbolic mysticism**: 72-year cycle (deep cultural patterns)
- **Media savvy**: 8-year cycle (technological adoption waves)
- **Charismatic leadership**: 15-year cycle (leader emergence patterns)
- **Messianic leadership**: 45-year cycle (rare salvational moments)

These different periodicities create complex resonance patterns. The 2010s peak represents alignment of short-cycle dimensions (anti-establishment, populist appeal, media savvy) amplifying aggregate antagonism despite stability in long-cycle dimensions.

4.5 Bootstrap Validation Across Nine Dimensions

4.5.1 Comprehensive Nine-Dimensional Validation

The 1000-iteration bootstrap validation confirms statistical robustness across all nine dimensions and their interactions:

Metric	Mean	95% CI	Stability
Network Density	649	[0.621, 0.677]	0.0028
Average Chain Length	15.33	[14.20, 16.50]	0.0027
Clustering Coefficient	708	[0.663, 0.753]	0.0032
Number of Attractors	21	[19, 23]	0.0048

Economic Ideology Mean	512	[0.487, 0.537]	0.0024
Social Ideology Mean	498	[0.471, 0.525]	0.0027
Messianic Leadership Mean	554	[0.534, 0.600]	0.0029
Charismatic Leadership Mean	623	[0.592, 0.654]	0.0025
Anti-Establishment Mean	667	[0.649, 0.713]	0.0024
Symbolic Mysticism Mean	408	[0.313, 0.510]	0.0040
Populist Appeal Mean	667	[0.586, 0.736]	0.0033
Authoritarian Mean	590	[0.500, 0.678]	0.0019
Media Savvy Mean	487	[0.456, 0.518]	0.0032

All stability coefficients below 0.005 confirm excellent reliability. Symbolic mysticism shows highest variance but remains statistically stable.

4.5.2 Dimensional Correlation Stability

Bootstrap analysis of correlations between dimensions reveals stable relationships:

Strong positive correlations ($r > 0.5$):

- Populist appeal ↔ Anti-establishment ($r = 0.72$)
- Messianic leadership ↔ Symbolic mysticism ($r = 0.68$)
- Charismatic leadership ↔ Messianic leadership ($r = 0.61$)
- Authoritarian ↔ Messianic leadership ($r = 0.58$)

Strong negative correlations ($r < -0.5$):

- Economic state intervention ↔ Media savvy ($r = -0.54$)
- Institutional legitimacy ↔ Charismatic leadership ($r = -0.62$)

Independent dimensions ($|r| < 0.2$):

- Social ideology shows weak correlation with most other dimensions
- Media savvy operates independently of ideological dimensions

These stable correlations reveal the deep structure of political space—certain dimensional combinations naturally co-occur while others remain orthogonal.

4.6 The Corpus Size Effect in Nine Dimensions

4.6.1 Dimensional Revelation with Expanded Corpus

Comparison between 13 and 32 document analyses shows dramatic revelation of nine-dimensional complexity:

Dimensional Coverage	13 Documents	32 Documents	Change
Dimensions with >3 positions	4	9	125%
Total unique 9D positions	11	21	91%
Dimensional variance (mean)	82	156	90%
Empty regions in 9D space	67%	28%	-58%

The smaller corpus severely undersampled nine-dimensional space, with entire dimensions effectively collapsed. Only economic ideology, social ideology, populist appeal, and authoritarian tendency showed variation in the small corpus. The expanded corpus reveals variation across all nine dimensions, fundamentally changing the political landscape.

4.6.2 Minimum Data Requirements for Nine-Dimensional Analysis

Statistical power analysis reveals minimum document requirements for reliable nine-dimensional political analysis:

- **5 documents**: Can distinguish 2-3 dimensions, severe undersampling
- **10 documents**: Can identify 4-5 dimensions, major patterns visible
- **20 documents**: Can map 7-8 dimensions, most structure apparent
- **30 documents**: All 9 dimensions measurable, attractors stable
- **50+ documents**: Full nine-dimensional dynamics, fine structure visible

Our 32-document corpus barely exceeds the minimum for complete nine-dimensional analysis, explaining why the jump from 13 documents produced such dramatic revelations. Future studies should aim for 50+ documents to fully explore nine-dimensional political space.

5. Discussion

5.1 Theoretical Implications of Nine-Dimensional Political Space

5.1.1 Transcending Traditional Political Categories

The nine-dimensional framework fundamentally challenges traditional political categorization schemes. The conventional left-right spectrum captures only 20.8% of genealogical connections, while the remaining 79.2% operate through other dimensional channels—populist strategies, authority structures, symbolic frameworks, and communication innovations. This finding suggests that most political influence and evolution occurs outside

traditional ideological categories, in dimensions that political science has systematically underexamined.

The independence of certain dimensions—particularly social ideology's weak correlation with other dimensions—explains persistent puzzles in political analysis. Why do economically left and right movements sometimes converge on social conservatism? Why do democratic and authoritarian regimes sometimes employ identical symbolic mysticism? The nine-dimensional framework reveals these aren't paradoxes but natural consequences of dimensional independence.

The strong correlations between other dimensions—populist appeal with anti-establishment rhetoric, messianic with symbolic mysticism—identify natural dimensional clusters that form coherent political syndromes. These clusters explain why certain political configurations repeatedly emerge across different contexts. The "populist syndrome" (high populist appeal + anti-establishment + charismatic leadership) appears in left and right variants because it represents a stable dimensional configuration independent of economic ideology.

5.1.2 Political Evolution as Nine-Dimensional Navigation

Conceiving political evolution as navigation through nine-dimensional space transforms our understanding of political change. Political movements don't simply move left or right but navigate a complex landscape with multiple degrees of freedom. Success requires not just ideological positioning but coordinated movement across all nine dimensions.

The differential transmission fidelity across dimensions—high for symbolic mysticism and populist appeal, low for media savvy and charismatic leadership—reveals that political traditions maintain identity through selective dimensional conservation. Core dimensions preserve tradition while peripheral dimensions enable adaptation. This selective conservation explains how movements maintain recognizable identity across centuries despite surface transformations.

The three transmission patterns—preservation, drift, and jumping—represent different evolutionary strategies in nine-dimensional space. Preservation maintains successful configurations, drift enables gradual adaptation, and jumping allows rapid repositioning in response to crises. The coexistence of all three patterns ensures both stability and innovation in political evolution.

5.1.3 Attractors as Evolutionary Stable Strategies

The 21 attractors identified in nine-dimensional space can be understood as evolutionary stable strategies (ESS) in game-theoretic terms—configurations that, once established, resist invasion by alternative strategies. Each attractor represents a local optimum in the fitness landscape, offering advantages that make deviation costly.

The dimensional composition of attractors—some defined primarily by economic ideology, others by populist appeal or symbolic mysticism—reveals multiple viable strategies for political success. There's no single optimal position in nine-dimensional space but rather multiple stable configurations, each exploiting different dimensional combinations. This multiplicity explains political diversity as an equilibrium phenomenon rather than transitional confusion.

The distribution of attractors—large mainstream attractors surrounded by smaller experimental ones—suggests a core-periphery structure in political evolution. Core attractors represent proven strategies refined over generations, while peripheral attractors explore nove

l configurations that might become future cores or evolutionary dead-ends. This structure maintains both stability and innovation potential.

5.2 Methodological Contributions of Nine-Dimensional Analysis

5.2.1 Computational Advantages of Higher-Dimensional Analysis

The nine-dimensional framework provides computational advantages beyond theoretical insight. Higher-dimensional spaces better separate distinct political positions, reducing classification errors and revealing subtle differences. Documents that appear identical in low-dimensional projections separate clearly in nine dimensions, enabling precise genealogical tracing and influence detection.

The dimensional decomposition of connections, antagonisms, and cycles enables targeted analysis of specific political aspects. Researchers can examine economic genealogies independently of symbolic evolution, or trace populist innovations regardless of ideological content. This analytical flexibility surpasses holistic approaches that conflate different political dimensions.

The bootstrap validation across nine dimensions, achieving stability coefficients below 0.005, demonstrates that higher-dimensional analysis need not sacrifice statistical rigor. Indeed, the additional dimensions provide more information for validation, strengthening confidence in results. The stable dimensional correlations provide internal consistency checks unavailable in lower-dimensional analyses.

5.2.2 Revealing Hidden Political Patterns

The nine-dimensional analysis reveals patterns invisible to traditional methods:

Cross-ideological technical transfer: Documents share media savvy and anti-establishment rhetoric across ideological divides, explaining how opposition movements often mirror each other's tactics despite conflicting goals.

Symbolic persistence amid ideological change: High-fidelity transmission of symbolic mysticism explains why certain narratives ("civilization versus barbarism") persist despite complete ideological reversals.

Dimensional specialization of political actors: Some documents innovate primarily on single dimensions while maintaining conventional positions elsewhere, revealing a division of labor in political evolution.

Multidimensional resonance in crisis: The 2010s antagonism peak results from short-cycle dimensions aligning, showing how multiple political rhythms can amplify into systemic crisis.

These patterns, requiring nine-dimensional analysis to detect, suggest that much political behavior dismissed as irrational or contradictory actually follows comprehensible multidimensional logic.

5.2.3 Practical Applications for Political Analysis

The nine-dimensional framework enables practical applications:

Political positioning strategy: Movements can identify underexploited regions of nine-dimensional space, positioning themselves in attractive but unoccupied niches.

Coalition possibility analysis: Parties can identify potential allies who share positions on key dimensions despite diverging on others.

Opposition prediction: Governments can anticipate which dimensions opponents will emphasize based on current positioning and historical patterns.

Crisis early warning: Monitoring dimensional antagonisms and cycles enables prediction of high-risk periods when multiple dimensions align.

Message crafting: Communications can target specific dimensions knowing which have low transmission thresholds and high impact.

The computational tools make these applications accessible to practitioners, not just academic researchers.

5.3 Implications for Democratic Theory and Practice

5.3.1 Multidimensional Democratic Competition

The nine-dimensional framework reconceptualizes democratic competition. Rather than simple left-right contestation, democracy involves competition across multiple dimensions simultaneously. Parties compete not just on policy but on authority structures, symbolic narratives, communication strategies, and populist framing. This multidimensional

competition explains why democratic politics often appears chaotic—multiple games play out simultaneously in different dimensions.

The differential antagonism trends—intensifying on anti-establishment and populist dimensions while stable on others—suggests that democratic stress concentrates in specific dimensions rather than spreading uniformly. This concentration creates opportunities for targeted interventions. Rather than addressing "polarization" generally, democratic strengthening can focus on dimensions showing dangerous antagonism levels.

The dimensional cycles with different periodicities create a complex temporal landscape for democratic politics. Short cycles (6-year anti-establishment, 12-year populist) create frequent upheaval, while long cycles (72-year symbolic) provide underlying stability. Understanding these multiple temporalities helps distinguish surface turbulence from deep structural change.

5.3.2 Democratic Resilience in Nine Dimensions

Democratic resilience depends on maintaining viable positions across all nine dimensions, not just winning economic or social ideological debates. Democracies that neglect dimensions like symbolic mysticism or media savvy become vulnerable to challengers who exploit these neglected dimensions. The multidimensional nature of political competition means democratic defense requires multidimensional strategies.

The existence of 21 stable attractors suggests democracy isn't a single configuration but a region in nine-dimensional space containing multiple democratic attractors. Some emphasize economic equality, others individual liberty; some rely on charismatic leadership, others on institutional process; some invoke transcendent values, others remain secular. This multiplicity strengthens democracy by providing fallback positions—if one democratic attractor fails, others remain available.

The high network density (0.649) indicates that democratic and non-democratic positions interconnect extensively in nine-dimensional space. This connectivity enables both democratic breakdown (connections to authoritarian attractors) and democratic recovery (connections back to democratic attractors). The dense network means no position is permanently isolated—both hope and danger persist.

5.3.3 Designing Nine-Dimensional Democratic Interventions

Understanding nine-dimensional political space enables sophisticated democratic interventions:

Dimensional bridging: Creating new documents/speeches that connect democratic attractors across multiple dimensions, building coalitions previously thought impossible.

Attractor reinforcement: Strengthening democratic attractors by populating them with more documents, making them more attractive to political actors.

Dimension-specific depolarization: Targeting interventions at dimensions showing dangerous antagonism rather than attempting general depolarization.

Cycle disruption: Timing interventions to prevent resonance between different dimensional cycles that could amplify into crisis.

Strategic dimensional innovation: Introducing new positions on underutilized dimensions to reshape political competition away from polarizing dimensions.

These interventions require nine-dimensional thinking—understanding that changing one dimension affects others through correlation structures, that timing matters due to different cycles, and that success requires coordinated positioning across all dimensions.

5.4 Limitations and Future Directions

5.4.1 Limitations of Nine-Dimensional Framework

Despite its advantages, the nine-dimensional framework faces limitations:

Dimensional selection: While empirically grounded, the choice of nine specific dimensions remains partly subjective. Other researchers might identify different dimensions or subdivide existing ones. Future work should explore optimal dimensionality through information-theoretic criteria.

Static dimensionality: The framework assumes nine dimensions remain relevant across 215 years, but political evolution might create new dimensions or obsolete old ones. Adaptive dimensionality that allows dimensions to emerge and disappear could better capture political evolution.

Cultural specificity: The nine dimensions derive from Argentine political experience and might not transfer to other contexts. Cross-national studies should test whether these dimensions are universal or culturally specific.

Measurement challenges: Some dimensions, particularly symbolic mysticism and messianic leadership, rely on subjective interpretation of language that automated analysis might miss. Improved natural language processing could enhance measurement precision.

5.4.2 Extensions to Contemporary Politics

The framework requires extensions to analyze contemporary digital politics:

Social media dynamics: New dimensions might be needed to capture viral propagation, algorithmic amplification, and network effects specific to digital platforms.

Transnational dimensions: Globalization creates political connections transcending national boundaries, requiring dimensions that capture international positioning.

Algorithmic mediation: Political messages increasingly pass through recommendation algorithms that might create new evolutionary pressures and dimensional dynamics.

Multimodal politics: Contemporary politics involves images, videos, and memes that text analysis misses. Multimodal analysis could reveal additional dimensions.

5.4.3 Future Research Directions

The nine-dimensional framework opens multiple research directions:

Comparative dimensional analysis: Apply the framework to other countries to identify universal versus particular dimensions.

Real-time tracking: Develop systems to map contemporary political discourse in nine-dimensional space as it evolves.

Experimental validation: Test predicted dimensional dynamics through controlled experiments with political messages.

Normative implications: Explore which dimensional configurations promote democratic values versus authoritarian outcomes.

Intervention design: Develop and test nine-dimensional interventions to strengthen democracy.

Evolutionary modeling: Create agent-based models of political evolution in nine-dimensional space to predict future trajectories.

Dimensional reduction: Investigate whether certain dimensions can be combined without losing analytical power.

Causal inference: Move beyond correlation to identify causal relationships between dimensional positions and political outcomes.

6. Conclusions

6.1 Core Findings

This study demonstrates that political evolution operates through complex mechanisms best understood in nine-dimensional space. The analysis of Argentine political discourse from 1810 to 2025 reveals patterns invisible to traditional approaches:

The identification of 21 stable political attractors in nine-dimensional space shows that politics operates through more diverse configurations than conventional analysis suggests. These attractors, defined by distinct combinations across nine dimensions—economic ideology, social ideology, messianic leadership, charismatic leadership, anti-establishment rhetoric, symbolic mysticism, populist appeal, authoritarian tendency, and media savvy—represent evolutionary stable strategies that persist across regime changes and crises.

The 322 genealogical chains averaging 15.33 connections demonstrate multi-generational transmission of political ideas that transcend traditional ideological boundaries. Only 20.8% of connections operate through economic ideology, while the majority transmit through other dimensional channels—shared populist strategies, similar authority structures, common symbolic frameworks. This reveals that most political influence operates outside conventional left-right categories.

The monotonic increase in political antagonism (Historical: 0.436, Contemporary: 0.496) masks heterogeneous dimensional trends. Anti-establishment rhetoric and populist appeal drive polarization, while authoritarian tendency and social ideology remain stable. This dimensional decomposition enables targeted understanding of democratic stress rather than vague concerns about "polarization."

The dramatic difference between 13 and 32 document analyses—91% more attractors, 571% more genealogical chains—reveals that political complexity has been systematically underestimated due to insufficient data. The nine-dimensional space requires approximately 30 documents minimum to populate adequately, below which entire dimensions collapse and patterns vanish.

6.2 Theoretical Contributions

The study advances political science theory through multiple innovations:

The nine-dimensional framework transcends the limitations of traditional left-right spectrums, revealing that most political behavior operates in dimensions that conventional analysis ignores. Political movements navigate a complex nine-dimensional landscape where success requires coordinated positioning across all dimensions, not just ideological coherence.

The identification of dimension-specific transmission fidelity—high for symbolic mysticism and populist appeal, low for media savvy and charismatic leadership—explains how political traditions maintain identity while adapting. Core dimensions preserve continuity while peripheral dimensions enable innovation, resolving the paradox of simultaneous stability and change.

The discovery of three transmission patterns—preservation, drift, and jumping—reveals different evolutionary strategies in political space. These patterns explain how orthodox traditions, gradual adaptations, and revolutionary appropriations coexist within the same political system.

The attractors as evolutionary stable strategies provide game-theoretic foundations for political diversity. Multiple stable configurations exist in nine-dimensional space, each exploiting different dimensional combinations. This multiplicity explains political pluralism as equilibrium rather than transition.

6.3 Methodological Innovations

The computational tools developed—PoliticalRootFinder and PoliticalMemespace—enable unprecedented precision in political analysis:

Nine-dimensional semantic mapping reveals connections invisible to traditional methods. Documents that appear unrelated ideologically connect through shared dimensions like media savvy or symbolic mysticism. This multidimensional connectivity explains cross-ideological influence patterns that confound conventional analysis.

Bootstrap validation achieving stability coefficients below 0.005 across all nine dimensions demonstrates that higher-dimensional analysis need not sacrifice statistical rigor. The additional dimensions provide more information for validation, strengthening confidence beyond what lower-dimensional analyses achieve.

The dimensional decomposition of political phenomena—connections, antagonisms, cycles—enables targeted analysis of specific political aspects. Researchers can trace economic genealogies independently of symbolic evolution, or examine populist innovations regardless of ideological content.

6.4 Implications for Democratic Practice

The findings carry profound implications for understanding and strengthening democracy:

Democratic competition involves simultaneous contests across nine dimensions, not simple left-right opposition. Parties compete on authority structures, symbolic narratives, and communication strategies alongside policy positions. This multidimensional competition explains democratic complexity while revealing intervention opportunities.

Democratic resilience requires maintaining viable positions across all nine dimensions. Democracies that neglect certain dimensions become vulnerable to challengers who exploit these gaps. The 21 attractors suggest democracy encompasses multiple stable configurations rather than a single ideal type, providing flexibility and fallback positions.

The dimensional specificity of polarization—concentrating in anti-establishment and populist dimensions—enables targeted interventions rather than vague "depolarization" efforts. Understanding dimensional cycles prevents dangerous resonances that amplify into crises.

6.5 Future Directions

This study opens numerous research avenues:

Cross-national application of the nine-dimensional framework could identify universal versus culturally specific dimensions. Do all democracies exhibit nine dimensions? Do the same dimensions drive polarization globally?

Real-time tracking of contemporary politics in nine-dimensional space could provide early warning of democratic stress. Social media analysis could map political evolution as it occurs rather than retrospectively.

Experimental validation could test causal relationships between dimensional positions and political outcomes. Do certain dimensional configurations promote democratic stability? Can strategic positioning in nine-dimensional space strengthen democratic actors?

The development of nine-dimensional intervention strategies could translate analytical insights into democratic practice. How can democratic actors navigate nine-dimensional space strategically? What dimensional innovations might revitalize democratic competition?

6.6 Final Reflections

As democracies worldwide confront unprecedented challenges—authoritarian resurgence, technological disruption, environmental crisis, economic inequality—understanding political evolution becomes urgent. The nine-dimensional framework reveals that these challenges operate across multiple dimensions simultaneously, requiring multidimensional responses.

The Argentine case, with its dramatic oscillations and persistent antagonisms, provides lessons for all democracies. The survival of democratic attractors despite authoritarian interruptions offers hope. The multiplicity of stable configurations suggests diverse democratic futures remain possible. The genealogical connections across apparent ideological chasms reveal common ground obscured by surface conflicts.

Yet the framework also warns of dangers. The monotonic increase in antagonism, particularly in anti-establishment and populist dimensions, suggests systemic pressures toward polarization. The dimensional cycles create predictable crisis periods when multiple rhythms align. The dense network connectivity means democratic breakdown remains an ever-present possibility.

The nine-dimensional analysis transforms political understanding from flat ideological maps to rich topographical landscapes. Politics becomes navigation through complex terrain with multiple paths, dead ends, and hidden connections. This complexity might seem daunting, but it also reveals opportunities invisible to simpler analyses.

The computational tools now available enable unprecedented precision in political analysis. The nine dimensions can be measured, tracked, and predicted with statistical rigor. Yet tools

alone don't ensure wisdom. Understanding political evolution mechanisms creates possibilities for intervention but doesn't determine values guiding their use.

The mathematics of nine-dimensional political evolution reveals constraints and possibilities but not destinies. Attractors shape but don't determine trajectories. Genealogies influence but don't dictate positions. Antagonisms intensify but can be reversed. Cycles create rhythms but can be disrupted. Within mathematical constraints, human agency remains decisive.

The question is whether we will use these insights to navigate nine-dimensional political space toward democratic flourishing or drift toward breakdown. The dimensions are mapped. The attractors are identified. The cycles are calculated. The tools are available. The choice of direction remains ours—but now, at least, we have a compass showing all nine dimensions of the political landscape we must traverse.

Democracy's future depends not on winning a single ideological dimension but on successfully navigating all nine simultaneously. This complexity makes democratic politics challenging but also resilient—multiple dimensions provide multiple opportunities for democratic renewal. The nine-dimensional framework doesn't simplify politics but makes its true complexity comprehensible and, perhaps, manageable.

May we use this understanding wisely, strengthening democratic attractors across all dimensions while remaining alert to the multidimensional nature of political evolution. The path forward requires not just ideological clarity but nine-dimensional navigation skills. The framework provides the map; the journey remains ours to make.

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Data Availability Statement

All data and code available at: https://github.com/adrianlerer/peralta-metamorphosis

Expanded analysis results available in project repository.
Bootstrap validation datasets available upon request.

Appendices

Appendix A: Nine-Dimensional Measurement Specifications

Dimensional Definitions and Measurement

The nine dimensions of political space are measured through keyword analysis with contextual weighting and temporal calibration:

1. **Economic Ideology** (0=state intervention, 1=market orientation)
2. **Social Ideology** (0=progressive, 1=conservative)
3. **Messianic Leadership** (0=institutional, 1=salvational)
4. **Charismatic Leadership** (0=institutional legitimacy, 1=personal charisma)
5. **Anti-Establishment Rhetoric** (0=system-preserving, 1=system-challenging)
6. **Symbolic Mysticism** (0=secular/rational, 1=mystical/transcendent)
7. **Populist Appeal** (0=pluralist, 1=populist)
8. **Authoritarian Tendency** (0=democratic, 1=authoritarian)
9. **Media Savvy** (0=low adaptation, 1=high sophistication)

Each dimension employs curated keyword dictionaries, contextual analysis, and temporal calibration. For example, "liberal" in 1850 scores differently than "neoliberal" in 1990. Media savvy is calibrated to contemporary technology—radio mastery in 1940s equals social media expertise in 2020s.

Full measurement specifications and keyword dictionaries available at: https://github.com/adrianlerer/peralta-metamorphosis/docs/dimensions.md

Appendix B: Bootstrap Validation Summary

Bootstrap validation (1000 iterations) confirms statistical robustness:

Metric	Mean	95% CI	Stability Coefficient
Network Density	649	[0.621, 0.677]	0.0028
Average Chain Length	15.33	[14.20, 16.50]	0.0027
Number of Attractors	21	[19, 23]	0.0048

All nine dimensions show stability coefficients < 0.005, with symbolic mysticism showing highest variance (0.0040) but remaining statistically stable.

Complete bootstrap results and correlation matrices available at: https://github.com/adrianlerer/peralta-metamorphosis/results/bootstrap_validation.csv

Appendix C: Computational Implementation

The analysis employs two primary tools:

PoliticalRootFinder: Constructs genealogical networks through semantic similarity in nine-dimensional space, enforcing temporal directionality and calibrated similarity thresholds (τ=0.3).

PoliticalMemespace: Maps documents to nine-dimensional political space through sophisticated scoring algorithms accounting for context, negation, and temporal evolution.

Complete source code, documentation, and usage examples available at:

- GitHub: https://github.com/adrianlerer/peralta-metamorphosis
- Documentation: https://github.com/adrianlerer/peralta-metamorphosis/docs
- Interactive notebooks: https://github.com/adrianlerer/peralta-metamorphosis/notebooks

The repository includes:

- /src: Complete Python implementation of all algorithms
- /data: Preprocessed corpus of 32 documents
- /analysis: Jupyter notebooks reproducing all results
- /visualizations: Interactive network and dimensional visualizations
- /tests: Unit tests and validation scripts

Version used for this analysis: v1.0.0 (git commit: [hash])

Data Availability Statement

All data, code, and supplementary materials necessary to reproduce this analysis are publicly available:

Primary Repository: https://github.com/adrianlerer/peralta-metamorphosis

Contents:

- Complete source code for PoliticalRootFinder and PoliticalMemespace
- Preprocessed corpus of 32 Argentine political documents (1810-2025)
- Bootstrap validation results (1000 iterations)
- Interactive visualizations and network graphs
- Comprehensive documentation and usage examples

Licensing:

- Code: MIT License (open source, freely reusable)
- Data: Creative Commons CC-BY 4.0
- Analysis: Reproducible under same licenses

Requirements:

- Python 3.8+
- Dependencies listed in requirements.txt
- Computational requirements: 16GB RAM, ~2 hours for full analysis with bootstrap

For questions, collaboration, or assistance with reproduction: adrian@lerer.com.ar

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